BIKES

A Moving Network Instrument for Music and Sound Art

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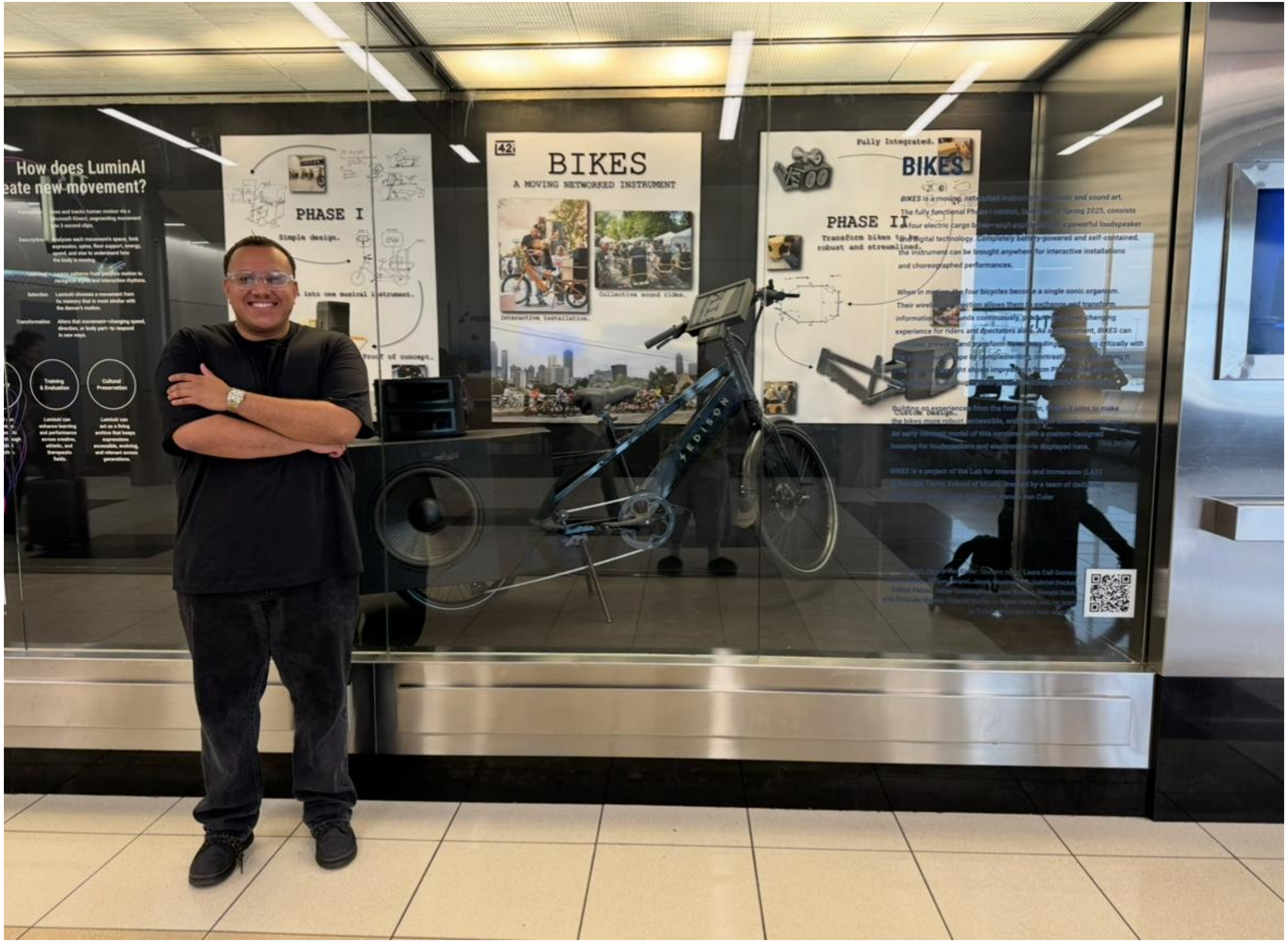
BIKES in one year:

- Hartsfield International Airport Exhibit
- SITE by Goat Farm
- Streets Alive
- Launchpad Design Showcase
- Inman Park Parade
- IEEE Internet of Sounds
- ICMC
- Undergraduate Research Symposium
- MOBB Group Ride


How does LuminAI
eate new movement?
PHASE I
Simple design.
BIKES
A MOVING NETWORKED INSTRUMENT
Interactive installation.
Collective sound rides.
Fully Integrated.
BIKES
PHASE II
Transform bikes to be
robust and streamlined.
EDISON

Recap

- BIKES is an interdisciplinary project, I am the **project lead**, leading a group of undergraduate students.
  - Six Students, Bi-Weekly Meetings since last Spring.
- Last semester we explored leveraging the network as a compositional parameter. (i.e. topologies)
- User feedback we kept getting - **How does distance affect a bike's sound?**

Research Question

How can **spatial relationships** between performers be a **compositional parameter** in a mobile networked performance system?

Hypothesis

Ultra-Wideband (UWB) sensors can enable **precise, real-time capture of spatial relationships** between bikes, allowing these relationships to serve as **meaningful compositional parameters** in a mobile networked performance system.

Related Work

- Urban sound can be structured through **soundscape theory and acoustic perception frameworks** (Schafer, 1977)
- **Urban environments** have been explored as **interactive musical systems** through **sensor-driven sound generation** *(Gaye et al., 2003)*
- **Movement** through space has been shown to structure **musical interaction and temporal flow** *(Rueb, 2004)*
- Bicycle-based systems have enabled **site-specific sonic experiences** in urban contexts *(Morinaga, 2018)*
- Environmental sensing has been used for **participatory and responsive sonic interaction** *(Griffiths et al., 2017)*
- Ultra-wideband (UWB) sensing enables **precise real-time distance measurement** *(Liu & Bao, 2021)*
- UWB has also been widely used for **localization** in **distributed sensor networks** *(Gezici et al., 2005)*

What is an Ultra-Wideband Sensor?

- UWB sensors are a short-range, low-power radio technology that uses **precise, high-frequency** pulses to detect exact location, distance, and movement of objects with **centimeter-level accuracy**. ToF technology.
- **Anchor vs. Tag model**
  - They send messages to each other.
  - You can be **either** a Tag or an Anchor at one instant in time.

Apple Airtag

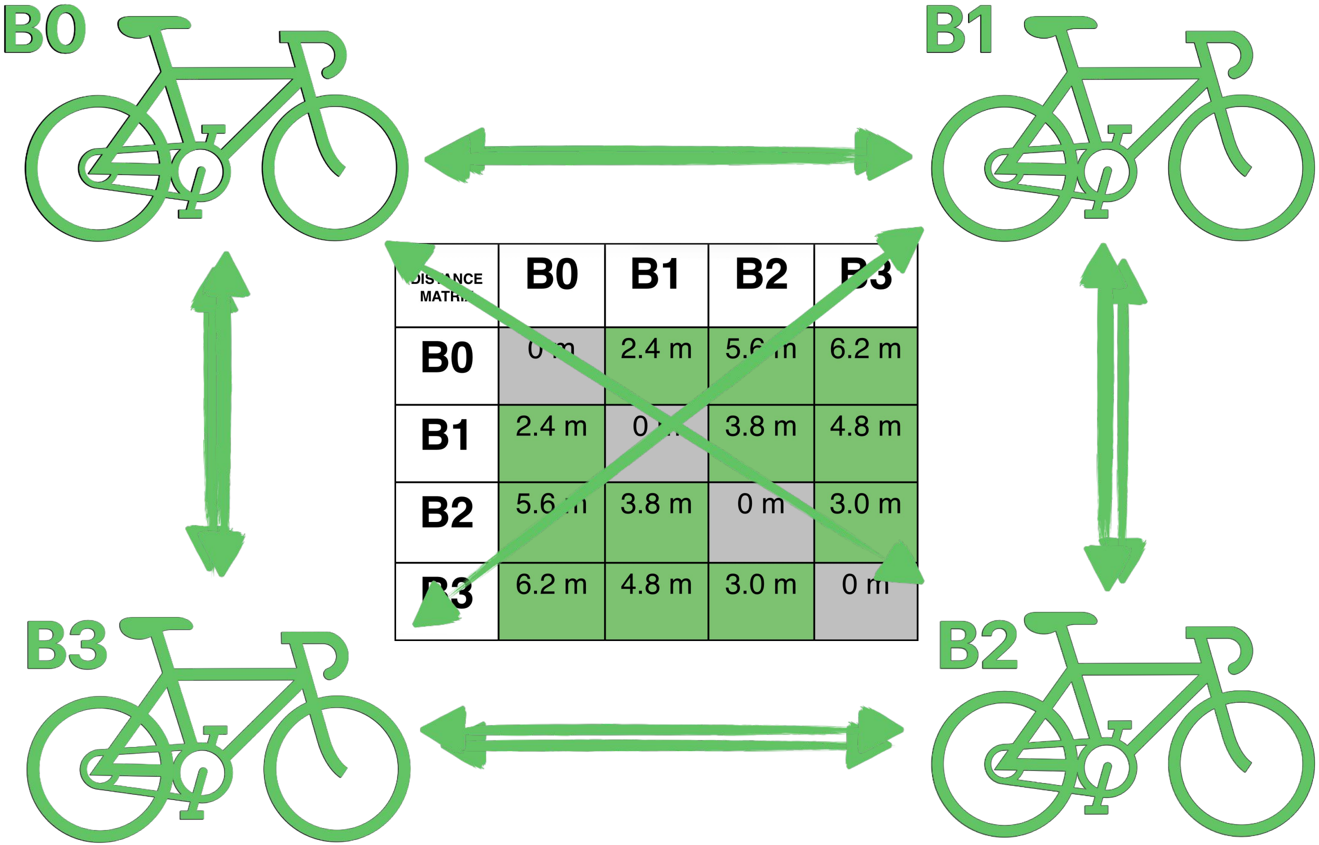
UWB Implementation onto BIKES

- In theory, we can use this technology to measure the **relative distance** between all bikes during a ride. (200M range, 100Hz)
- Hardware - The sensors are small [2.5 x 1 inch] and low power. (Build enclosure, Power/Data via Pi).
- Software- Leveraging the **Anchor vs. Tag model**... A bit more to consider.

ESP 32 UWB Sensor

BIKES UWB Relative Distance Algorithm

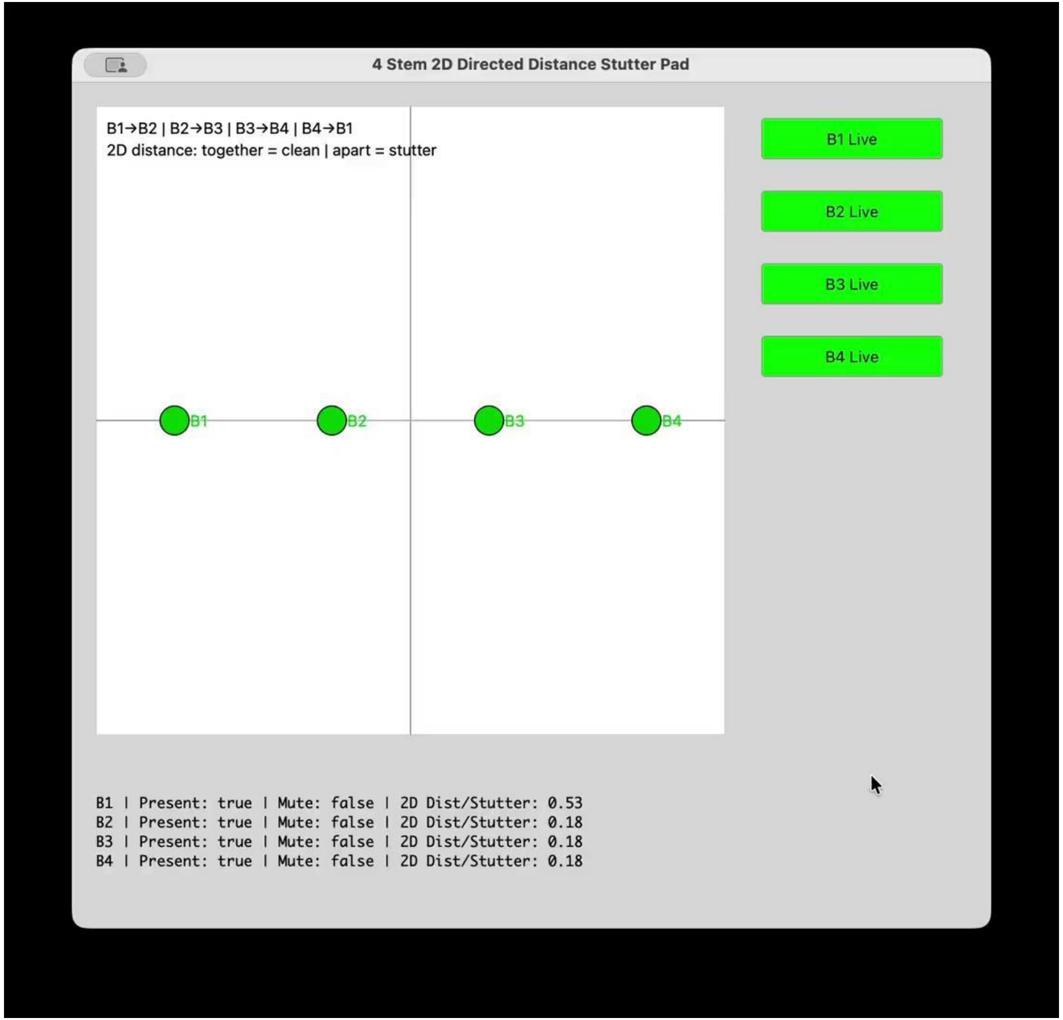
- Capture the distances between all BIKES
  - Need to work around our Tag vs. Anchor limitation
- A combination of Round Robin + State Machine (~30Hz, **really 10-20Hz**)
  - Round Robin = **Scheduling Order** of Tag vs. Anchor roles.
  - State Machine = **Control over switching** between roles..



DISTANCE MATRIX	B0	B1	B2	B3
B0	0 m	2.4 m	5.6 m	6.2 m
B1	2.4 m	0 m	3.8 m	4.8 m
B2	5.6 m	3.8 m	0 m	3.0 m
B3	6.2 m	4.8 m	3.0 m	0 m

Using relative distance as a compositional parameter

- UWB Data -> Serial Port -> Pi -> SuperCollider
- With a complete distance matrix, we can start to think about geometry.
- Now we have **spatial relationships as a compositional parameter.**
- **In my next demo, sound is modulated by a BIKE's proximity to its neighbor.**


4 Stem 2D Directed Distance Stutter Pad
B1→B2 | B2→B3 | B3→B4 | B4→B1
2D distance: together = clean | apart = stutter
B1
B2
B3
B4
B1 Live
B2 Live
B3 Live
B4 Live
B1 | Present: true | Mute: false | 2D Dist/Stutter: 0.53
B2 | Present: true | Mute: false | 2D Dist/Stutter: 0.18
B3 | Present: true | Mute: false | 2D Dist/Stutter: 0.18
B4 | Present: true | Mute: false | 2D Dist/Stutter: 0.18

References

Gaye, L., Mazé, R., & Holmquist, L. E. (2003). Sonic city: The urban environment as a musical interface. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)* (pp. 109–115).

Rueb, T. (2004). On the road: The drive project and locative media. In *Proceedings of the International Symposium on Electronic Art (ISEA)* (pp. 45–48).

Morinaga, G. (2018). Women who build things: Gestural controllers, augmented instruments, and musical mechatronics. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*.

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Schafer, R. M. (1977). *The soundscape: Our sonic environment and the tuning of the world*. Destiny Books.

Liu, Y., & Bao, Y. (2021). Real-time remote measurement of distance using ultra-wideband (UWB) sensors. *Automation in Construction*, *123*, 103569. https://doi.org/10.1016/j.autcon.2020.103569

Gezici, S., Tian, Z., Giannakis, G. B., Kobayashi, H., Molisch, A. F., Poor, H. V., & Sahinoglu, Z. (2005). Localization via ultra-wideband radios: A look at positioning aspects for future sensor networks. *IEEE Signal Processing Magazine*, *22*(4), 70–84. https://doi.org/10.1109/MSP.2005.1458289

Come experience the integration of spatial relationships into BIKES on May 6th between 2:30 - 4:30 PM outside of Couch!

One more note...

Thank You!